UAV sources

Local flight state
Position / velocity / heading

Private credits
Byte budget / credit balance

Protected attempts and retries
Bounded retries / failure recovery

Reports

Controls

Intermittent wireless service

Uplink
Data / registration / release

Future service rights
Owner / slot / capacity / expiry

Downlink
Grants / ACKs

Paid bytes: data + control

Received reports

Service events

Service rights

Grants / ACKs

Ground monitoring platform

State monitor
Reassembly / estimation of sensing state

History

H(t)

Causal resource support
Exact no-report marginals / resource allocation

Supports

Executable admission and ACK guard
Remaining obligations / ACK consistency

Quality bounds

Valid quality profile
Valid model bounds / Paid registration

Outstanding ACK debt (generated, not yet transmitted)